Light

GCSE Computer Science – Programming Fundamentals (Set 1)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is the difference between a variable and a constant in programming?

- ○ **A.** A variable stores text while a constant stores numbers
- ○ **B.** A variable can change value during program execution but a constant cannot change once set
- ○ **C.** A constant can change value during program execution but a variable cannot change once set
- ○ **D.** A variable is always global while a constant is always local

Q2 (1 mark)

Which data type would be most appropriate to store the number of students in a class?

- ○ **A.** Real
- ○ **B.** String
- ○ **C.** Integer
- ○ **D.** Boolean

Q3 (1 mark)

Which data type should be used to store a temperature reading of 36.8 degrees?

- ○ **A.** Integer
- ○ **B.** Boolean
- ○ **C.** Character
- ○ **D.** Real

Q4 (1 mark)

Which data type should be used to store whether a user has agreed to terms and conditions?

- ○ **A.** Boolean
- ○ **B.** Integer
- ○ **C.** String
- ○ **D.** Real

Q5 (1 mark)

What is the result of 17 MOD 5?

- ○ **A.** 3
- ○ **B.** 2
- ○ **C.** 3.4
- ○ **D.** 85

Q6 (1 mark)

What is the result of 29 DIV 6?

- ○ **A.** 5
- ○ **B.** 4.83
- ○ **C.** 4
- ○ **D.** 5.83

Q7 (1 mark)

What is the programming construct called where instructions are executed one after another in order?

- ○ **A.** Iteration
- ○ **B.** Selection
- ○ **C.** Sequence

- ○ **D.** Recursion

Q8 (1 mark)

Which programming construct allows a program to make decisions based on conditions?

- ○ **A.** Sequence
- ○ **B.** Selection
- ○ **C.** Iteration
- ○ **D.** Assignment

Q9 (1 mark)

What is the programming construct called that repeats a section of code multiple times?

- ○ **A.** Selection
- ○ **B.** Sequence
- ○ **C.** Declaration
- ○ **D.** Iteration

Q10 (1 mark)

Which comparison operator checks if two values are equal?

- ○ **A.** =

- B. !=
- C. ==
- D. <>

Q11 (1 mark)

What is the purpose of casting in programming?

- A. To convert a value from one data type to another data type
- B. To permanently delete a variable from the program
- C. To create a backup copy of a variable
- D. To encrypt data for security purposes

Q12 (1 mark)

What is the result of True AND False?

- A. True
- B. False
- C. Error
- D. Null

Q13 (2 marks)

A program needs to store a student's full name, their percentage score as a decimal, and whether they passed. Which combination of data types is most appropriate?

- ○ **A.** String, Integer, Boolean
- ○ **B.** String, Real, Boolean
- ○ **C.** Integer, Real, String
- ○ **D.** Real, String, Boolean

Q14 (2 marks)

What is the value of the following expression?

```
8 + 4 * 3 - 2
```

- ○ **A.** 18
- ○ **B.** 34
- ○ **C.** 12
- ○ **D.** 38

Q15 (2 marks)

How many times will the message be printed?

```
for i = 1 to 6
    print("Hello")
next i
```

- ○ **A.** 5

- B. 6
- C. 7
- D. 1

Q16 (2 marks)

What type of iteration is shown in this code?

```
while total < 100
    total = total + 10
endwhile
```

- A. Count-controlled iteration
- B. Condition-controlled iteration
- C. Nested iteration
- D. Infinite iteration

Q17 (2 marks)

What is output by this code?

```
first = "Computer"
second = "Science"
result = first + " " + second
print(result)
```

- A. ComputerScience
- B. Computer Science

- ○ **C.** first second
- ○ **D.** Computer + Science

Q18 (2 marks)

Given the array declaration:

```
array colours[4]
colours = ["Red", "Green", "Blue", "Yellow"]
```

What is the value of colours[2]?

- ○ **A.** "Red"
- ○ **B.** "Green"
- ○ **C.** "Blue"
- ○ **D.** "Yellow"

Q19 (2 marks)

What is the main difference between a function and a procedure?

- ○ **A.** A function returns a value to the calling code, a procedure does not return a value
- ○ **B.** A procedure can take parameters but a function cannot
- ○ **C.** A function can only perform calculations while a procedure can only display output
- ○ **D.** A procedure runs faster than a function

Q20 (2 marks)

What does it mean when a variable is described as 'local'?

- ○ **A.** The variable can be accessed from any part of the program
- ○ **B.** The variable is stored on the local computer rather than a server
- ○ **C.** The variable can only be accessed within the subroutine where it was declared
- ○ **D.** The variable stores location data like postcodes

Q21 (2 marks)

What is the result of the expression?

```
(10 > 15) OR (8 < 12)
```

- ○ **A.** True
- ○ **B.** False
- ○ **C.** 10
- ○ **D.** 8

Q22 (2 marks)

What is the value of result?

```
word = "ALGORITHM"
result = word.left(4)
```

- A. "ALGO"
- B. "RITHM"
- C. "ALGOR"
- D. "A"

Q23 (2 marks)

What does the following code produce?

```
number = random(1, 10)
```

- A. Always returns the value 10
- B. Returns a random integer between 1 and 10 inclusive
- C. Returns either 1 or 10 only
- D. Returns a random decimal number

Q24 (3 marks)

What is output by this code?

```
score = 75
if score >= 80 then
    grade = "A"
elseif score >= 60 then
    grade = "B"
elseif score >= 40 then
    grade = "C"
else
    grade = "D"
```

```
endif
print(grade)
```

- A. A
- B. B
- C. C
- D. D

Q25 (3 marks)

What is the value of result?

```
x = 47
y = 8
result = (x DIV y) + (x MOD y)
```

- A. 15
- B. 12
- C. 5
- D. 13

Q26 (3 marks)

A program needs to read data from an existing file and display it. Which sequence of file operations is correct?

- A. Open, Write, Close, Read
- B. Read, Open, Close

- **C.** Open, Read, Close
- **D.** Close, Open, Read

Q27 (3 marks)

Which SQL statement correctly retrieves all customers from London with a balance over 500?

- **A.** SELECT Customers FROM * WHERE London AND 500
- **B.** GET * FROM Customers IF city = "London" AND balance > 500
- **C.** FIND Customers WHERE city = "London" OR balance > 500
- **D.** SELECT * FROM Customers WHERE city = "London" AND balance > 500

Q28 (3 marks)

A 2D array is declared as:

```
array grid[5,4]
```

How many elements can this array store, and what is the index of the last element?

- **A.** 20 elements, last index is [4,3]
- **B.** 9 elements, last index is [5,4]
- **C.** 20 elements, last index is [5,4]
- **D.** 25 elements, last index is [4,4]

Q29 (3 marks)

What is output by this code?

```
function calculate(a, b)
    result = (a + b) * 2
    return result
endfunction

value = calculate(5, 3) + calculate(2, 4)
print(value)
```

- ○ **A.** 32
- ○ **B.** 16
- ○ **C.** 14
- ○ **D.** 28

Q30 (3 marks)

What is the value of result?

```
a = 15
b = 20
c = 10
result = NOT(a > b) AND (b > c)
```

- ○ **A.** 20
- ○ **B.** False
- ○ **C.** 15
- ○ **D.** True

Total: **55** marks • Time: **30 min**

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